EL-MOASSER

By a group of supervisors

Pure Mathematics Problems

- Algebra & Analytic Solid Geometry
- Differential & Integral Calculus



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جميع حقوق الطبع والنشر محفوظة

لا يجوز، بأى صورة من الصور، التوصيل (النقل) المباشر أو غير المباشر لأى مما ورد فى هذا الكتاب أو نسخه أو تصويره أو ترجمته أو تحويره أو الاقتباس منه أو تحويله رقميًا أو إتاحته عبر شبكة الإنترنت **إلا بإذن كتابى** مسبق من الناشر.

كما لا يجوز بأى صورة من الصور استخدام العلامة التجارية (EL-MOASSER) المسجلة باسم الناشر

ومَن يخالف ذلك يتعرض للمساءلة القانونية طبقًا لأحكام القانون ٨٢ لسنة ٢٠٠٢ الخاص بحماية الملكية الفكرية.

Preface

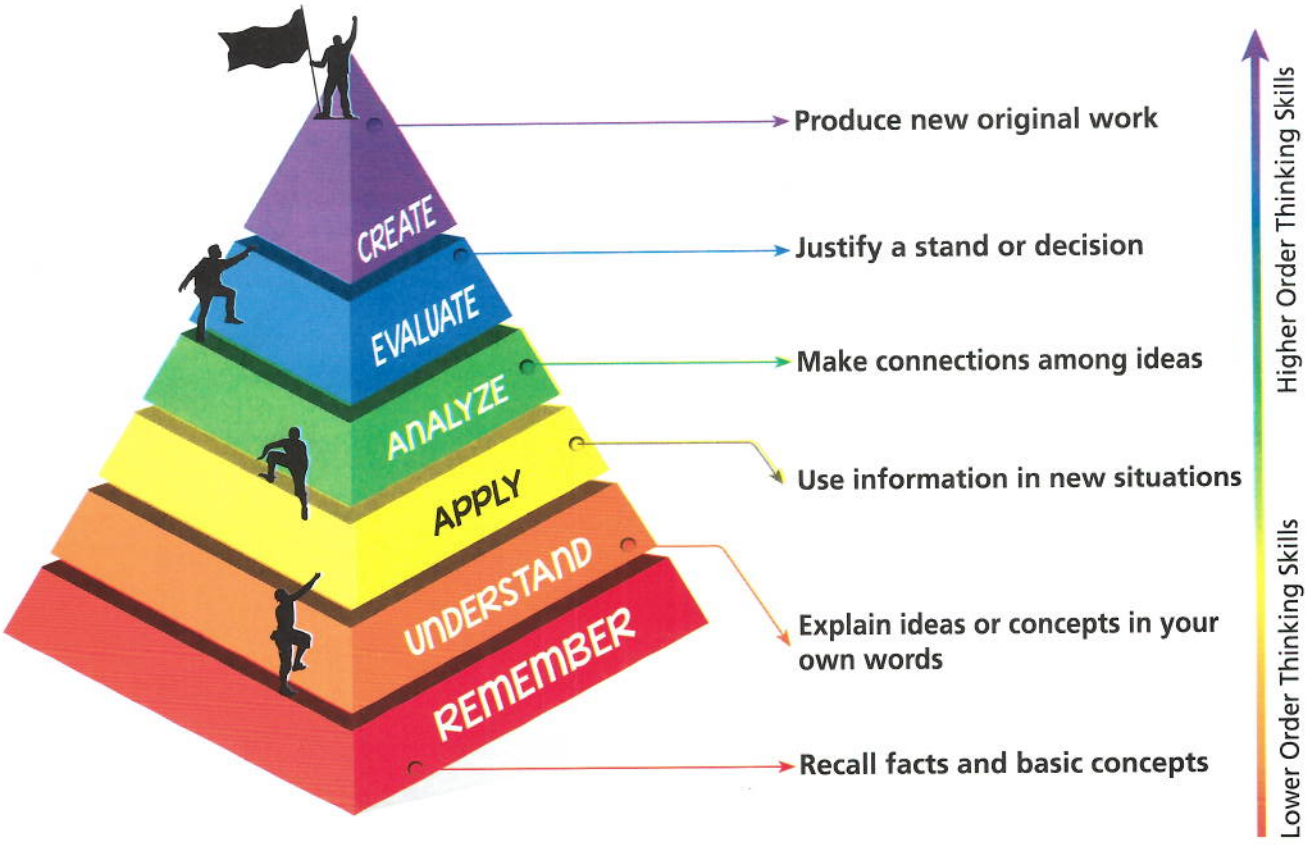
Thanks to God who helped us to introduce one of our famous series
"El Moasser" in mathematics.
We introduce this book to our colleagues.
We also introduce it to our students to help them study mathematics.
In fact, this book is the outcome of more than thirty years experience in the field of teaching mathematics.
This book will make students aware of all types of questions.
We would like to know your opinions about the book hoping that it will win your admiration.
We will be grateful if you send us your recommendations and your comments.

The Authors

Bloom's Taxonomy Of Cognitive Levels

Bloom's Taxonomy is an educational classification created by Benjamin Bloom, it is often represented as a pyramid. This taxonomy was revised to include six cognitive levels graded form the lower level to the higher level as follows:

REMEMBER → UNDERSTAND → APPLY → ANALYZE → EVALUATE → CREATE



Bloom's Revised Pyramid

Note :

The questions within each exercise are classified according to the levels of Bloom's pyramid and are referred to as follows:

● UNDERSTAND ○ APPLY Higher Order Thinking Skills (HOTS)
(ANALYZE - EVALUATE - CREATE)

CONTENTS

First | Algebra and Analytic Solid Geometry

Unit 1 Binomial theorem

Unit 2 Complex numbers

Unit 3 Geometry and measurement in three dimensions

Unit 4 Straight lines and planes in space

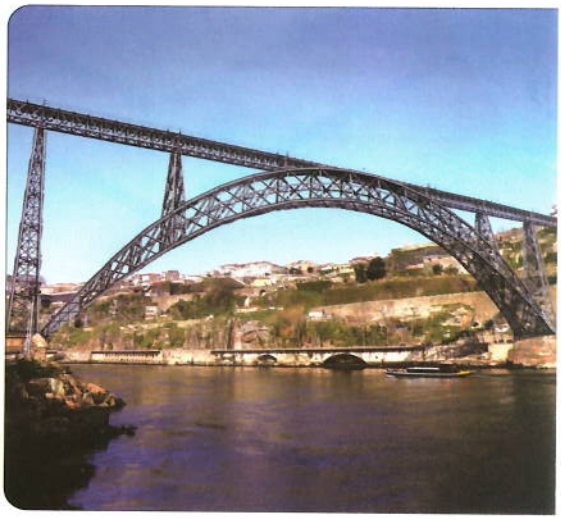


Second | Differential and Integral Calculus

Unit 1 Differentiation and its applications.

Unit 2 Behavior of the function and curve sketching.

Unit 3 The definite integral and its applications.



First

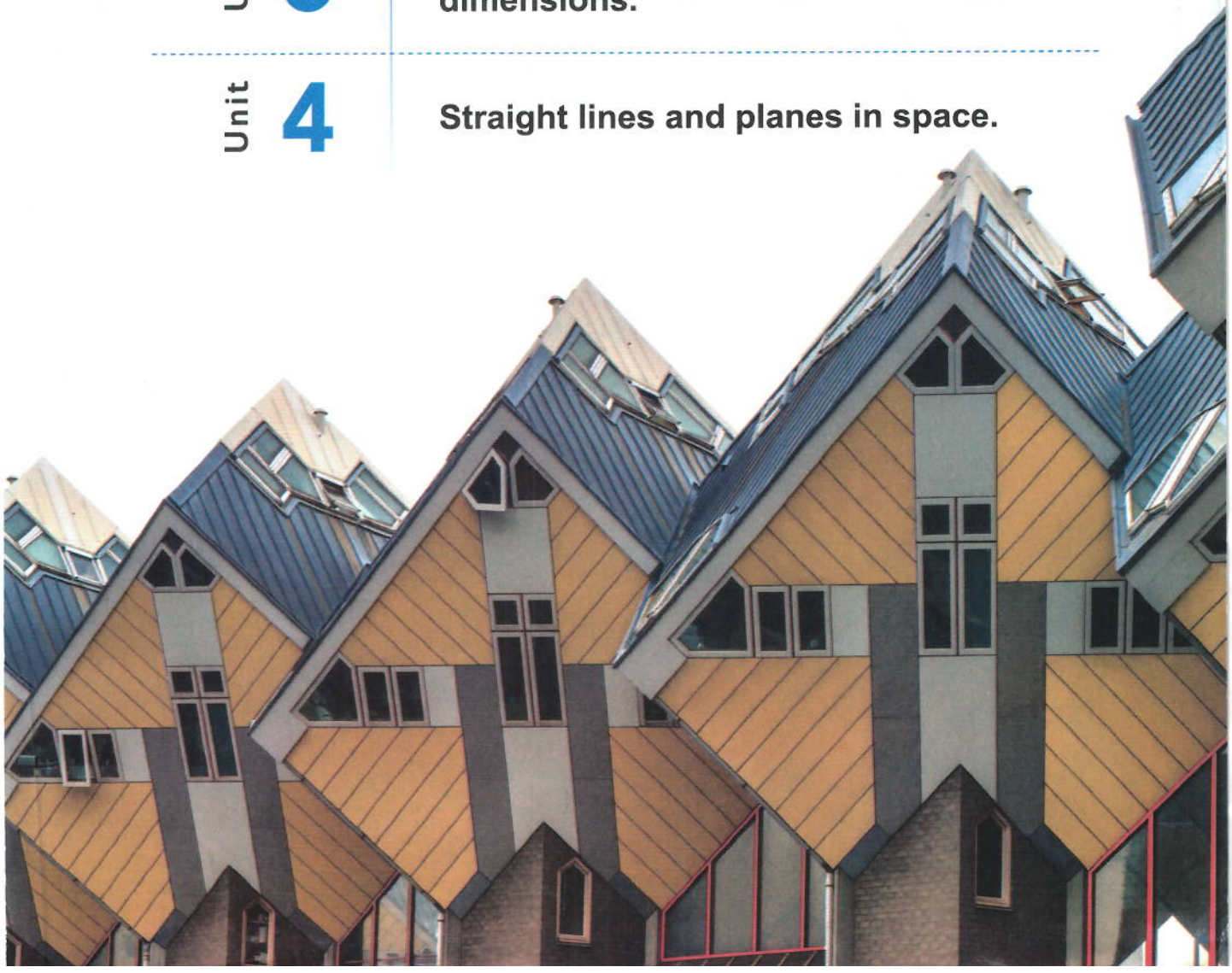
Algebra and Analytic Solid Geometry

Unit 1 Binomial theorem.

Unit 2 Complex numbers.

Unit 3 Geometry and measurement in three dimensions.

Unit 4 Straight lines and planes in space.



UNIT

ONE

Binomial theorem

Exercise **1** Binomial theorem for positive integer power.

Exercise **2** Finding the term containing x^r in the expansion of binomial.

Exercise **3** Ratio between two consecutive terms in the binomial expansion.



Exercise 1

On binomial theorem for positive integer power

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **By using the binomial theorem , find the expansion of the following :**

(1) $(x+y)^5$

(2) $(x-2)^6$

(3) $(2x+3y)^4$

(4) $\left(x-\frac{1}{x}\right)^5$

(5) $\left(2x^2-\frac{3}{2x}\right)^5$

(6) $(1-x)^5(1+x)^5$

2 **Write in the simplest form the expansion of each of the following :**

(1) $(x+2)^6-(x-2)^6$

(2) $\left(x+\sqrt{2}\right)^4+\left(x-\sqrt{2}\right)^4$

(3) $\left(1+\sqrt{x}\right)^5-\left(1-\sqrt{x}\right)^5$ and find the value of the result when $x=3$

(4) $\left(\frac{x^3+1}{x+1}+x\right)^4+(x^2-1)^4$

3 **Write the expansion for each of :**

(1) $(1-x)^8$, then use it to find the value of :

$1-{}^8C_1+{}^8C_2-{}^8C_3+\cdots+{}^8C_8$

(2) $(a+2b)^6$ from this prove that :

${}^6C_1+{}^6C_2+{}^6C_3+{}^6C_4+{}^6C_5+{}^6C_6=63$

(3) $(1+x)^n$ and then prove that :

(1) ${}^nC_0+{}^nC_1+{}^nC_2+\cdots+{}^nC_n=2^n$

(2) ${}^8C_1+2\times{}^8C_2+3\times{}^8C_3+\cdots+8\times{}^8C_8=8\times2^7$

4 **Choose the correct answer from those given :**

(1) If the middle term in the expansion of : $\left(\frac{2a}{3}+\frac{b}{a^2}\right)^{8n}$ is the ninth term , then n = ...

(a) 1 (b) 2 (c) 3 (d) 4

(2) The fourth term in the expansion of : $\left(x+\frac{1}{x}\right)^4$ according to the descending order of the exponents of x equals .....

(a) $4x^2$ (b) $\left(\frac{1}{x}\right)^4$ (c) $\frac{1}{x^2}$ (d) $\frac{4}{x^2}$

(3) ***(2nd Session 2023)*** The number of terms of the expansion of $(x^2-6x+9)^{21}$ after simplifying equals ..... term.

(a) 43 (b) 42 (c) 22 (d) 21

(4) The number of terms in the expansion of $\left((a+4b)^3(a-4b)^3\right)^2$ equals .....

(a) 6 (b) 7 (c) 8 (d) 32

(5) If the number of terms in the expansion of : $(x+y)^{2n-1}$ equals 12 terms , then n = ...

(a) 5 (b) 6 (c) 7 (d) 12

(6) If the number of terms in the expansion of : $(a+b)^{2n}+(a-b)^{2n}$ equals 11 , then n = ...

(a) 8 (b) 9 (c) 10 (d) 11

(7) 📖 In the expansion of : $(1+bx)^9$ according to the ascending exponents of x , the coefficient of the sixth term is .....

(a) ${}^9C_5$ (b) ${}^9C_6$ (c) ${}^9C_5\,b^5$ (d) ${}^9C_6\,b^6$

(8) The coefficient of the middle term in the expansion of : $\left(3x-\frac{1}{6}\right)^{10}$ equals .....

(a) $\frac{-63}{8}$ (b) $\frac{-67}{8}$ (c) $\frac{63}{8}$ (d) $\frac{67}{8}$

(9) ***(1st Session 2022)*** The sum of the coefficients of the two middle terms in the expansion of $\left(\frac{2}{x}+\frac{x^2}{2}\right)^{25}$ equals .....

(a) $2\,{}^{25}C_{13}$ (b) $4\,{}^{25}C_{12}$ (c) $\frac{5}{2}\,{}^{25}C_{12}$ (d) $\frac{2}{5}\,{}^{25}C_{13}$

(10) ***(2nd Session 2022)*** The coefficient of the third term in the expansion of $\left(x^2+ax+\frac{a^2}{4}\right)^4$ after simplifying according to the descending exponents of x is ..... where $a \in \mathbb{Z}$

(a) $7a^2$ (b) $\frac{1}{7}a^2$ (c) $\frac{3}{2}a^2$ (d) $\frac{2}{3}a^2$

(11) The sum of coefficients of the terms of the expansion : $(1 - 2x^2)^{2025}$ = ...

(a) – 1 (b) 1 (c) 0 (d) 2025

(12) The sum of the coefficients of the terms of the expansion : $(5x + 3y)^n$ equals 64 , then n = ...

(a) 2 (b) 3 (c) 6 (d) 8

(13) If the sum of the terms coefficients in the expansion of $(a^2x^2 - 2ax + 1)^9$ equals zero , then a = ...

(a) 2 (b) – 2 (c) 1 (d) – 1

(14) If a is the sum of the coefficients of the odd-order terms in the expansion of $\left(2x - \frac{3}{x^2}\right)^{19}$ and b is the sum of the coefficients of the even-order terms in the same expansion , then a + b = ...

(a) – 1 (b) zero (c) 1 (d) 5

(15) In the expansion of a binomial , it has 7 positive terms and 6 negative terms , then the expression is in the form of : .....

(a) $(a - b)^{12}$ (b) $(a + b)^{13}$ (c) $(a + b)^{12}$ (d) $(a - b)^{13}$

(16) ***(1st Session 2019)*** The last term in the expansion of : $(2 - x)^5 (2 + x)^5$ according to the ascending exponents of x is .....

(a) x^5 (b) $-x^5$ (c) $-x^{10}$ (d) x^{10}

(17) In the expansion of : $(1 + ax)^7$ according to the ascending exponents of x , if the coefficient of $T_5 = 560$, then a = ...

(a) 2 (b) 4 (c) ± 2 (d) ± 4

(18) ***(Trial 2021)*** If the coefficient of the ninth term in the expansion of $\left(a\sqrt{x} - \frac{1}{a\sqrt{x}}\right)^{12}$ according to the descending exponents of x equals 7920 , then a = ...

(a) $\pm \frac{1}{2}$ (b) ± 2 (c) $\pm \frac{1}{4}$ (d) ± 4

(19) ***(Trial 2021)*** In the expansion of $(x + a)^n$ according to the descending exponents of x , if T_4 is the same as the fifteenth term from the end , then n = ...

(a) 16 (b) 17 (c) 18 (d) 19

(20) If the coefficients of the sixth and sixteenth terms in the expansion of : $(x + y)^n$ are equals , then n = ...

(a) 19 (b) 20 (c) 21 (d) 22

(21) If the middle term in the expansion of : $\left(3x^2 + \frac{2}{3x}\right)^8$ equals 17920 , then $x =$ .....

(a) ± 2 (b) 3 (c) ± 4 (d) 5

(22) If T_{15} is the middle term in the expansion of $\left(a\sqrt{x} - \frac{1}{a\sqrt{x}}\right)^{n+3}$ according to the descending exponents of x , then coefficient of T_{16} = ... when $a = \frac{1}{2}$

(a) $-4 \times {}^{28}C_{13}$ (b) $4 \times {}^{28}C_{13}$ (c) $-4 \times {}^{26}C_{15}$ (d) $4 \times {}^{26}C_{15}$

(23) In the expansion $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$ if the ratio between the seventh term from the beginning to the seventh term from the end 1 : 6 , then n equals .....

(a) 7 (b) 8 (c) 9 (d) 10

(24) *(1st Session 2022)* If the ratio between the fourth term from the beginning to the fourth term from the end in the expansion of $\left(\frac{a}{x} + \frac{x}{a}\right)^{2n}$ equals $a^4 : x^4$, then n = ... (where $x \in \mathbb{R}^*$, $a \in \mathbb{R}^*$)

(a) 2 (b) 4 (c) 6 (d) 8

(25) *(2nd Session 2024)* If the two middle terms in the expansion of $(1 + x^2)^7$ are equal , then x^2 = ...

(a) $\frac{3}{5}$ (b) 1 (c) $\frac{5}{3}$ (d) 2

(26) In the expansion of : $(ax + b)^{2n+1}$, if the two middle terms are equal at $x = 2$, then .....

(a) $a = 2b$ (b) $b = 2a$ (c) $ab = 2$ (d) $ab = \frac{1}{2}$

(27) In the expansion of $(1 + x)^{17}$ according to the ascending exponents of x , if the coefficient of T_{r+4} is equal to the coefficient of T_{2r+3} , then r = ...

(a) 3 (b) 4 (c) 17 (d) 7

(28) In the expansion of $(1 + x)^n$, if the coefficient of T_b = coefficient of T_c , then b + c = ...

(a) n (b) n + 2 (c) n – 2 (d) 2n

(29) *(2nd Session 2022)* If the two middle terms in the expansion of $(1 + x)^{99}(1 - x + x^2)^{99}$ are equal , then x = ... where $x \in \mathbb{R}$

(a) 2 (b) 1 (c) – 2 (d) – 1

(30) If the coefficient of the middle term in the expansion of $(1 + mx)^4$ equals the coefficient of the middle term in the expansion of $(1 - mx)^6$, then m = ...

(a) $\frac{-3}{10}$ (b) $\frac{3}{10}$ (c) $\frac{-3}{5}$ (d) $\frac{3}{5}$

(31) ***(1st Session 2021)*** If the coefficient of T_6 in the expansion of $\left(a\,x + \frac{1}{b\,x}\right)^{10}$ according to the descending exponents of x equals ${}^{10}C_5$, then $\frac{a}{b}$ = ... where $a \in \mathbb{R}^*$, $b \in \mathbb{R}^*$

(a) -1 (b) 1 (c) 10 (d) $\frac{1}{10}$

(32) ***(Speciman 2025)*** In the expansion of $\left(x^2 + \frac{m}{x}\right)^{10}$ according to the descending exponents of x , if the middle term equals $\frac{7}{216}$, then $m\,x$ = ...

(a) 6 (b) $\frac{2}{3}$ (c) $\frac{1}{6}$ (d) $\frac{\sqrt{7}}{2}$

(33) In the expansion of : $(1 + x)^n = 1 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots + a_n x^n$ and $\frac{a_2 + a_3}{a_2} = 3$, then n = ...

(a) 4 (b) 6 (c) 8 (d) 9

(34) If $1 + \frac{5}{2}x + \frac{5 \times 4}{4\lfloor 2}x^2 + \frac{5 \times 4 \times 3}{8\lfloor 3}x^3 + \cdots + \frac{1}{32}x^5 = 1024$, then x = ...

(a) 5 (b) 4 (c) 6 (d) 8

(35) The set of solution of the equation :

$1 - 6x + \frac{6 \times 5}{2 \times 1}x^2 - \frac{6 \times 5 \times 4}{3 \times 2 \times 1}x^3 + \cdots + x^6 = 64$ in $\mathbb{R}$ is .....

(a) $\{-3 , 1\}$ (b) $\{1 , 3\}$ (c) $\{-1 , 3\}$ (d) $\{-1 , -3\}$

(36) Number of integer terms in the expansion $\left(1 + \sqrt{3}\right)^6$ before simplifying is .....

(a) 3 (b) 4 (c) 5 (d) 6

(37) Number of integer terms in the expansion $\left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)^7$ is .....

(a) zero (b) 2 (c) 3 (d) 4

(38) Number of terms of the expansion $(x + y)^{2000} + (x - y)^{2000}$ after reducing is .....

(a) 1000 (b) 2000 (c) 2001 (d) 1001

(39) Number of terms in the expansion $(x + y)^{1000} - (x - y)^{1000}$ after reducing is .....

(a) 1000 (b) 500 (c) 501 (d) 1001

(40) Number of terms in the expansion $(x + y)^{16} + (x - y)^{14}$ after reducing is .....

(a) 15 (b) 17 (c) 30 (d) 32

(41) If the number of terms of the expansion $(x + y)^n + (x - y)^n$ after reducing is 16 , then n = ...

(a) 32 only. (b) 31 only. (c) 30 or 31 (d) 31 or 32

(42) In the expansion of $\left(x + \frac{1}{x}\right)^{2n}$, the middle term ≠ ..... for every n > 1

(a) ${}^{2n}C_n$ (b) $\frac{{}^{2n}P_n}{\lfloor n}$

(c) ${}^{2n}P_n$ (d) $\frac{1 \times 3 \times 5 \times \cdots \times (2n - 1) \times 2^n}{\lfloor n}$

(43) $\sum_{r=0}^{20} {}^{20}C_r$ = ...

(a) 2^{20} (b) 2^{19}

(c) $2^{19} + \frac{1}{2}\,{}^{20}C_{10}$ (d) $2^{19} - \frac{1}{2}\,{}^{20}C_{10}$

(44) $\sum_{r=0}^{13} {}^{27}C_r$ = ...

(a) 2^{13} (b) 2^{14} (c) 2^{26} (d) 2^{27}

(45) If $(1 + a x)^n = 1 + 8x + 24x^2 + \cdots + a^n x^n$, then $\frac{a - n}{a + n}$ = ...

(a) 3 (b) – 3 (c) $-\frac{1}{3}$ (d) $\frac{1}{3}$

(46) $1 + {}^{20}C_1\, i + {}^{20}C_2\, i^2 + \cdots + {}^{20}C_{20}\, i^{20}$ = ...

(a) 20 (b) $\lfloor 20$ (c) – 1024 (d) 1024

(47) ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \cdots (-1)^n\, {}^nC_n$ = ...

(a) 2^n (b) $(-1)^n$ (c) zero (d) $(-2)^n$

(48) ${}^nC_0 + {}^nC_1 \left(\frac{1}{3}\right) + {}^nC_2 \left(\frac{1}{3}\right)^2 + {}^nC_3 \left(\frac{1}{3}\right)^3 + \cdots + {}^nC_n \left(\frac{1}{3}\right)^n$ = ...

(a) $\left(\frac{1}{3}\right)^n$ (b) $\left(\frac{2}{3}\right)^n$ (c) $\left(\frac{4}{3}\right)^n$ (d) $\left(\frac{5}{3}\right)^n$

(49) Number of terms in the expansions of

$\left[(x + y)^{10} + {}^{10}C_1\,(x + y)^9\,(x - y) + {}^{10}C_2\,(x + y)^8\,(x - y)^2 + \cdots + (x - y)^{10}\right]$

after reducing is .....

(a) 1 (b) 10 (c) 11 (d) 110

5 **Find :**

(1) The coefficient of the sixth term in the expansion of : $\left(x+\frac{2}{x}\right)^8$ according to the descending exponents of x

(2) The sixth term in the expansion of : $(2x-y)^7$ according to the ascending exponents of x

(3) The fifth term from the end of the expansion of : $\left(x^2-\frac{1}{x}\right)^{20}$ according to the descending exponents of x

(4) The middle term in the expansion of : $\left(2x+\frac{1}{2x^2}\right)^{12}$

(5) The two middle terms in the expansion of : $\left(\frac{x^2}{3}+\frac{3}{x}\right)^{15}$

(6) The fourth term from the end of the expansion of : $x^4\left(x-\frac{1}{x}\right)^9$ according to the descending exponents of x

(7) The middle term in the expansion : $1+{}^{12}C_1\,x+{}^{12}C_2\,x^2+\cdots+{}^{12}C_{12}\,x^{12}$

(8) The fifth term in the expansion :
$(3+x)^{11}-{}^{11}C_1\,(3+x)^{10}(1-2x)+{}^{11}C_2\,(3+x)^9(1-2x)^2-\cdots-(1-2x)^{11}$ after simplifying according to the ascending exponents of x

(9) The numerical value of the sixth term when : $x=1$ in the expansion :
$(1-x)^8+24x(1-x)^7+252x^2(1-x)^6+\cdots+6561x^8$ after simplifying according to the ascending exponents of x

(10) The middle term in the expansion of : $(3+2x)^8+(3-2x)^8$

6 **Find the real value of x in each of the following if :**

(1) $1+8x+{}^8C_2\,x^2+\cdots+x^8=256$

(2) $\left(1+\sqrt{3}\right)^6-\left(1-\sqrt{3}\right)^6=480\sqrt{3}\,x$

(3) $\left(\frac{1}{2}x\right)^{10}-10\times3\left(\frac{1}{2}x\right)^9+45\times3^2\left(\frac{1}{2}x\right)^8-120\times3^3\left(\frac{1}{2}x\right)^7+\cdots+3^{10}=1024$

(4) $1+20x+190x^2+1140x^3+\cdots+x^{20}$
$=x^{18}+18x^{17}+153x^{16}+\cdots+1$

(5) The middle term in the expansion of : $\left(x^2+\frac{1}{2x}\right)^{10}$ equals $\frac{28}{27}$

(6) ***(1st Session 2015)*** The two middle terms in the expansion of : $\left(\frac{2}{x}+\frac{x^3}{8}\right)^9$ are equal.

(7) ***(Speciman 2025)*** The two middle terms in the expansion of :
$(2+x)^7+7x^2(2+x)^6+\frac{7\times6}{2\times1}x^4(2+x)^5+\cdots+x^{14}$ are equal.

(8) The sum of the two middle terms of the expansion : $\left(x+\frac{1}{x^2}\right)^9$ equals zero.

(9) ***(Sudan 2014)*** The third term in the expansion of : $\left(2x^2+\frac{1}{2}\right)^7$ according to the descending exponents of x is equal to the sixth term.

(10) The middle term from the expansion of : $(1+x)^{10}$ according to the ascending exponents of x equals the double of the seventh term.

(11) The sum of the three terms (first , middle and the last term) in the expansion of : $(x-1)^6$ equals 190

7 Answer the following questions :

(1) If the two middle terms in the expansion of : $(3x+2y)^{13}$ are equal.
Prove that : $\frac{x}{y}=\frac{2}{3}$

(2) In the expansion of $(1+mx)^n$, if the coefficients of x , x^2 are 12 , 63 respectively , what is the value of : m , n ?

(3) In the expansion of : $(1+x)^n$ according to the ascending exponents of x , if the coefficient of the sixth term equals the coefficient of the tenth term , find the value of : n

(4) In the expansion of : $(1+ax)^8$ according to ascending exponents of x if $T_2=4$ and the coefficient of $T_5=1120$, find the values of : a , x

(5) In the expansion of : $(1+x)^n$ according to the ascending exponents of x if $T_3=28x^2$, $T_5=1120$, find the value of n and x

(6) In the expansion of : $(1+kx)^{10}$ according to the ascending exponents of x , if the coefficient of the third term equals 180 , and the fifth term equals 210 , find the values of k and x

(7) In the expansion of $(1+x)^n$ according to the ascending exponents of x if $5(T_6)^2=18\,T_4\times T_8$, find the value of : n where n is a positive integer.

(8) From the expansion of : $(a\,x + b)^{10}$ according to the descending exponents of x , if the coefficient of $T_6 = \frac{63}{8}$, prove that : $2\,a\,b = 1$

(9) If a , b are the two middle terms in the expansion of : $\left(x - \frac{1}{x}\right)^{15}$ according to the descending exponents of x , prove that : $a + b\,x^2 = 0$

(10) If the coefficients of the 4th , 5th and 6th terms of the expansion $\left(2\,x + y\right)^n$ according to the descending exponents of x respectively , form an arithmetic terms , find value of n

(11) In the expansion of : $(1 + m\,x)^n$ according to the ascending exponents of x , if the first three terms are 1 , $\frac{-10}{3}\,x$ and $5\,x^2$ respectively , find the value of m , n then find the value of the middle term of this expansion at $x = 3$

(12) If the coefficient of T_4 in the expansion of : $(1 + x)^n$ according to the ascending exponents of x equals three times the coefficient of T_3 in the expansion of : $(1 + y)^{n-1}$ according to the ascending exponents of y , what is the value of n ?

(13) If the two middle terms in the expansion of : $\left(2\,a + \frac{1}{x}\right)^{2n+1}$ are equal and if $a + x = \frac{9}{4}$, find the value of each of : a and x

8 Answer the following :

(1) If $(m + x)^n = 3\,a + 6\,a\,x + 5\,a\,x^2 + \cdots + \in x^n$

Where : $n \in \mathbb{Z}^+$, find the value of each of : m , n and a

(2) If $(1 + m\,x)^{10}$ according to the ascending exponents of x equals

$a + b\,x + C\,x^2 + 15\,x^3 + \cdots + m^{10}x^{10}$, find the value of each of : m , a , b , c

(3) If $(1 + C\,x)^n = 1 + 20\,x + a_1\,x^2 + a_2\,x^3 + \cdots + a_{n-1}\,x^n$ and $16\,a_1 = 3\,a_2$, find the value of : n , C where $C \neq 0$

(4) If n is a positive integer and

$(1 + k\,x)^n = 1 + m_1\,x + m_2\,x^2 + m_3\,x^3 + m_4\,x^4 + \cdots + m_n\,x^n$

and $m_1 = 12$, $m_4 = 4\,m_2$, find the value of each of : n , k

(5) *(1st Session 2013)* If $(a - x)^{14} = b + b_1\,x + b_2\,x^2 + b_3\,x^3 + \cdots + b_{14}\,x^{14}$

and if $4\,b_4 + 11\,(b_3 + b_2) = 0$, prove that : $a = 2$

(6) If $(1 + x)^n = a_0 + a_1\,x + a_2\,x^2 + \cdots + a_n\,x^n$

Prove that : (1) $\frac{a_1}{a_0} + 2 \times \frac{a_2}{a_1} + 3 \times \frac{a_3}{a_2} + \cdots + n\,\frac{a_n}{a_{n-1}} = \frac{n}{2}\,(n + 1)$

(2) $a_0 + 2\,a_1 + 4\,a_2 + 8\,a_3 + \cdots + 2^n\,a_n = 3^n$

9 Prove that the coefficient of the middle term in the expansion of : $(x + y)^{2n}$ equals the sum of the coefficients of the two middle terms in the expansion of : $(x + y)^{2n-1}$

10 In the expansion of : $(a + b)^n$, if l was the sum of the odd ordered terms , m is the sum of the even ordered terms , **prove that :**

(1) $\ell^2 - m^2 = (a^2 - b^2)^n$ **(2)** $4\ell m = (a + b)^{2n} - (a - b)^{2n}$

11 **Prove that :**

(1) The expression $3^{2n} - 8n - 1$ is divisible by 64 by using the binomial theorem.

(2) The expression $3^{3n} - 1$ is divisible by 26 by using the binomial theorem.

12 **Prove that :** $\frac{{}^nC_r}{{}^{n-1}C_{r-1}} = \frac{n}{r}$

If the ratio between T_6 in the expansion of : $\left(x + \frac{1}{x}\right)^{15}$ according to the descending exponents of x , and T_5 in the expansion of $\left(x - \frac{1}{x^2}\right)^{14}$ according to the descending exponents of x equals $\frac{8}{9}$, find the value of x

Higher skills

13 **Choose the correct answer from those given :**

(1) ***(Speciman 2025)*** The coefficient of the middle term in the expansion of $(1 + 3x + 3x^2 + x^3)^4$ equals .....

(a) ${}^4C_2$ (b) ${}^4C_3$ (c) ${}^{12}C_6$ (d) ${}^{12}C_7$

(2) If $x > 1$ and T_3 in the expansion of : $\left(\frac{1}{x} + x^{\log x}\right)^5$ equals 1000 , then $x =$ .....

(a) 5 (b) 10 (c) $5\sqrt{2}$ (d) 100

(3) The number of integers terms in the expansion of : $\left(\sqrt{3} + \sqrt[8]{5}\right)^{256}$ before simplifying is .....

(a) 32 (b) 33 (c) 34 (d) 35

(4) ***(Speciman 2025)*** If $(1 + x + x^2)^9 = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_{18} x^{18}$, then $a_0 + a_2 + a_4 + \ldots + a_{18} =$ .....

(a) 365 (b) 2094 (c) 3281 (d) 9842

(5) The coefficient of T_{13} in the expansion of : $\sum_{r=0}^{50} {}^{50}C_r (x - 1)^{50-r} \times 2^r$ equals .....

(a) ${}^{50}C_{13}$ (b) ${}^{50}C_{12}$ (c) 1 (d) zero

(6) If $i^2 = -1$, $x + y\,i = \left(\sqrt{3} + \frac{i}{2}\right)^5 + \left(\sqrt{3} - \frac{i}{2}\right)^5$, then .....

(a) $y = 0$ (b) $x = 0$ (c) $x > 0$, $y > 0$ (d) $x > 0$, $y < 0$

(7) ${}^{10}C_0 + \frac{1}{2}\,{}^{10}C_1 + \frac{1}{3}\,{}^{10}C_2 + \frac{1}{4}\,{}^{10}C_3 + \cdots + \frac{1}{11}\,{}^{10}C_{10} =$ .....

(a) 2^9 (b) 2^{55} (c) $\left[\frac{1}{11}(2)^{11} - 1\right]$ (d) $\frac{3}{10}(2)^{10}$

(8) In the expansion of $(1 + x)^{59}$ according to the ascending exponents of x , sum of coefficients of the last 30 terms = ...

(a) 2^{29} (b) 2^{28} (c) 2^{58} (d) 2^{59}

(9) If the middle term in the expansion of : $\left(\frac{1}{x} + x \sin x\right)^{10}$ equals $7\frac{7}{8}$, then $x =$ .....

(a) $2n\pi + (-1)^n \times \frac{\pi}{6}$ (b) $n\pi + (-1)^n \times \frac{\pi}{6}$

(c) $n\pi + \frac{\pi}{6}$ (d) $n\pi + (-1)^n \times \frac{\pi}{3}$

14 **Prove that :** The middle term in the expansion of : $(1 + x)^{2n}$ equals

$$\frac{1 \times 3 \times 5 \times \cdots (2n-1)}{1 \times 2 \times 3 \times \cdots \times n}(2x)^n$$

15 **Prove that :**

(1) ${}^{25}C_0\,{}^{25}C_{10} + {}^{25}C_1\,{}^{25}C_{11} + {}^{25}C_2\,{}^{25}C_{12} + \cdots + {}^{25}C_{15}\,{}^{25}C_{25} = {}^{50}C_{15}$

(2) ${}^{25}C_0\,{}^{25}C_5 - {}^{25}C_1\,{}^{25}C_6 + {}^{25}C_2\,{}^{25}C_7 - \cdots + {}^{25}C_{20}\,{}^{25}C_{25} = {}^{25}C_{10}$